ResBridge: $SE(3)$ Self-Regulating Heteroscedastic Bridges for RNA Ensemble Generation

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Abstract

The abstract paragraph should be indented ½ inch (3 picas) on both the left- and right-hand margins. Use 10 point type, with a vertical spacing (leading) of 11 points. The word **Abstract** must be centered, bold, and in point size 12. Two line spaces precede the abstract. The abstract must be limited to one paragraph.

1 Introduction

Please read the instructions below carefully and follow them faithfully.

2 Background

2.1 RNA 3D Structure Modeling

2.2 Flow Matching

2.3 Flexibility Modeling

3 Methodology

We present **ResBridge**, a heteroscedastic stochastic bridge on $SE(3)^L$ for RNA backbone ensemble generation. Below we introduce the problem setup, the deterministic SE(3) flow matching backbone, the ResBridge framework itself, and its B-factor-guided variant ResBridge-guided.

3.1 Problem Definition

Given an RNA sequence s of length L, we aim to learn a conditional distribution $p_\theta(G \mid s)$ over backbone conformations whose i.i.d. samples $\{G^{(k)}\}_{k=1}^K$ satisfy two requirements: (i) accuracy — physically valid, native-like backbone and (ii) multi-modal coverage — the ensemble covers multiple conformational states. Following prior RNA backbone modeling Anand et al. [2025], each backbone is represented as a sequence of rigid frames, one per nucleotide, $G = \{(x_i, R_i)\}_{i=1}^L \in SE(3)^L$. Here $SE(3) = \mathbb{R}^3 \rtimes SO(3)$ is the Lie group of 3D rigid-body motions, each element of which pairs a translation with a rotation: $x_i \in \mathbb{R}^3$ is the coordinate of the frame origin (the $C4'$ atom) and $R_i \in SO(3) \subset SE(3)$ is the local backbone orientation defined by the three frame atoms $C4'$, $O4'$, and $C3'$. The full RNA backbone thus lives on the product manifold $SE(3)^L$.

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3.2 SE(3) Flow Matching Backbone

Flow matching Lipman et al. [2022], Liu et al. [2022] learns a neural velocity field that continuously transports a tractable prior p_0 to the data distribution p_{data} along a probability path $\{p_t\}_{t\in[0,1]}$. Since we represent RNA backbones as tuples of rigid frames in $SE(3)^L$, we instantiate flow matching directly on this manifold following Yim et al. [2023a], Bose et al. [2023]: the velocity field $v_\theta : SE(3)^L \times [0,1] \to \mathfrak{se}(3)^L$ outputs, at each residue frame $(x_t, R_t) \in SE(3)$, a translational component in $\mathbb{R}^3$ and a rotational component in $T_{R_t}SO(3) \cong \mathfrak{so}(3)$.

For each sample pair $(G_0, G_1) \sim p_0 \times p_{\text{data}}$ and time $t \sim \mathcal{U}(0,1)$, we take the conditional path to be the constant-speed geodesic on $SE(3)$ under its bi-invariant metric; differentiating it in t yields the closed-form target velocity

$$u_t^x = \frac{x_1 - x_t}{1-t}, \qquad u_t^R = \frac{\log(R_t^\top R_1)}{1-t} \in T_{R_t}SO(3). \tag{1}$$

Letting $\hat{v}_t^x$ and $\hat{v}_t^R$ denote the two heads of v_θ that predict these targets, we train by tangent-space regression

$$\mathcal{L}_{\text{SE(3)}}(\theta) \;=\; \mathbb{E}_{t,\,G_0,\,G_1}\Big[\lambda_x \big\|\hat{v}_t^x - u_t^x\big\|_2^2 \;+\; \lambda_R \big\|\hat{v}_t^R - u_t^R\big\|_{\text{F}}^2\Big], \tag{2}$$

with $\lambda_x, \lambda_R > 0$ balancing the two modalities and $\|\cdot\|_{\text{F}}$ the Frobenius norm on $\mathfrak{so}(3)$.

3.3 ResBridge: a tangent-space-lifted stochastic bridge

The deterministic SE(3) flow matching backbone of §3.2 transports each starting frame configuration $G_0 \sim p_0$ to a single endpoint via an ODE. Under multi-modal data this commits the velocity field to the conditional mean whenever several ground-truth endpoints are consistent with the same (G_t, s), and the noise-free sampler offers no second mechanism to disperse trajectories across distinct basins once one has been committed to. **ResBridge** addresses both routes by replacing the deterministic conditional path of §3.2 with a *stochastic bridge* on $SE(3)^L$ that is closed-form, endpoint-strict, and admits an analytically tractable drift, while leaving the loss, the network, and the velocity target of §3.2 entirely unchanged.

Three properties. We need a conditional distribution $p_t(G_t \,|\, G_0, G_1)$ that satisfies: (i) *closed-form sampling*, so that training remains simulation-free; (ii) *strict endpoint constraints*, $G_{t=0} = G_0$ and $G_{t=1} = G_1$ almost surely; (iii) *analytic drift*, so that the conditional flow matching loss of §3.2 can supervise the same velocity field on the same network. Translations are easy — the standard Euclidean Brownian bridge supplies all three. Rotations are not: the intrinsic Brownian bridge on $SO(3)$ generated by the Laplace–Beltrami operator has no closed-form simulator, since its transition density is the heat kernel and is only available as a series of Wigner D-functions, which is what motivates the truncated IGSO(3) approximations used by prior $SO(3)$ diffusion models Leach et al. [2022], Yim et al. [2023b]. We sidestep this difficulty by *lifting the bridge to the tangent space at R_0 and pushing forward through the exponential map*.

Tangent-space-lifted bridge on $SE(3)^L$. Per residue i, ResBridge defines the conditional sample $G_{i,t} = (x_{i,t}, R_{i,t})$ by two independent Brownian bridges, one in $\mathbb{R}^3$ and one in $\mathfrak{so}(3) \cong \mathbb{R}^3$. The translation channel is the Euclidean bridge

$$x_{i,t} \;=\; (1-t)\, x_{i,0} + t\, x_{i,1} + \sigma_x \sqrt{t(1-t)}\, \varepsilon_i^x, \qquad \varepsilon_i^x \sim \mathcal{N}(0, I_3), \tag{3}$$

with constant width $\sigma_x > 0$ and closed-form marginal $x_{i,t} \mid x_{i,0}, x_{i,1} \sim \mathcal{N}\big((1-t)x_{i,0} + t\, x_{i,1},\, \sigma_x^2\, t(1-t)\, I_3\big)$. The rotation channel takes the relative rotation $R_{i,0} \to R_{i,1}$ in the principal log

$$\omega_i \;=\; \log\big(R_{i,0}^\top R_{i,1}\big)^\vee \;\in\; \mathbb{R}^3 \;\cong\; \mathfrak{so}(3), \qquad |\omega_i| < \pi, \tag{4}$$

runs an Euclidean Brownian bridge on ω_i with constant width $\sigma_R > 0$,

$$\Omega_{i,t} \;=\; t\,\omega_i + \sigma_R \sqrt{t(1-t)}\, \varepsilon_i^R, \qquad \varepsilon_i^R \sim \mathcal{N}(0, I_3), \tag{5}$$

and pushes forward to $SO(3)$ via the right action of the exponential map at $R_{i,0}$,

$$R_{i,t} \;=\; R_{i,0}\, \exp\big(\widehat{\Omega_{i,t}}\,\big). \tag{6}$$

The pair (3)–(6) defines the ResBridge conditional distribution on $SE(3)^L$, parameterized by two constant scalar widths: σ_x in Å and σ_R in radians.

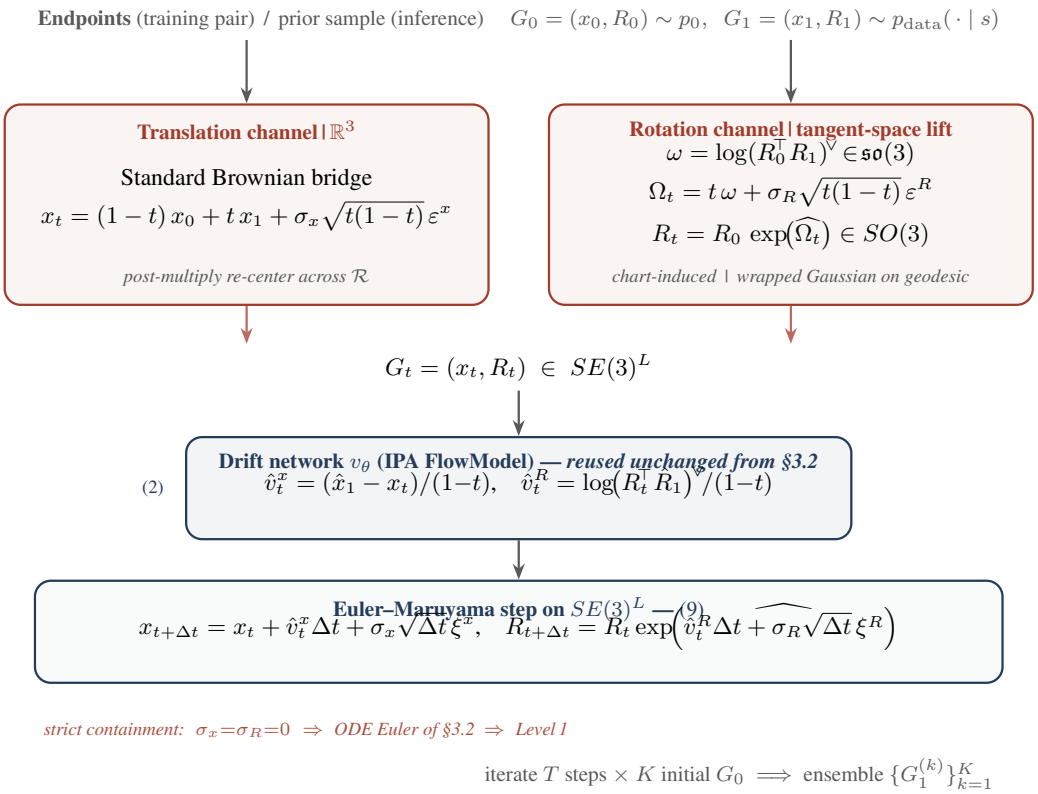

Figure 1: **ResBridge framework (Level 2).** ResBridge replaces the deterministic conditional path of §3.2 with two closed-form Brownian bridges (red), one in $\mathbb{R}^3$ for translations and one in the tangent space $\mathfrak{so}(3) \cong \mathbb{R}^3$ for rotations, the latter pushed back to $SO(3)$ through the exponential map at R_0. The corrupted state $G_t = (x_t, R_t)$ is fed into the same drift network v_θ (blue) as in the deterministic backbone of §3.2, supervised by the same flow matching loss (2); the only inference change is to replace the ODE Euler step by the Euler–Maruyama step (9). ResBridge therefore introduces *no new trainable parameter and no new loss term*, and strictly contains the deterministic backbone in the limit $\sigma_x = \sigma_R = 0$.

Honest framing. Because (5) is a Gaussian in a chart anchored at $R_{i,0}$, the resulting marginal $p_t(R_{i,t} \mid R_{i,0}, R_{i,1})$ is a *wrapped Gaussian around the geodesic*, not the heat kernel of $SO(3)$; a different choice of base point would give a different conditional law. ResBridge is therefore a **chart-induced** stochastic bridge rather than an intrinsic Brownian bridge, and we adopt this construction explicitly because it gives Properties (i)–(iii) in closed form. The IGSO(3)-based training prior of §3.2 keeps $|\omega_i|$ well below the cut locus π at every residue and timestep, so the chart artefact never manifests during training; the same holds at inference because the Euler-Maruyama loop only ever exponentiates a small per-step increment, never a full ω. We discuss this trade-off explicitly in the limitations (§3.4).

Endpoints, conditional law, and drift. The construction (3)–(6) admits the following two propositions, proved in Appendix A.

Proposition 1 (Endpoint constraint). Almost surely, $G_{i,0} = (x_{i,0}, R_{i,0})$ and $G_{i,1} = (x_{i,1}, R_{i,1})$ for every residue i.

Proposition 2 (Closed-form conditional law). Conditionally on the endpoints, the per-residue marginals factor and are Gaussians in their respective charts:

$$x_{i,t} \,|\, x_{i,0}, x_{i,1} \sim \mathcal{N}\big((1-t)x_{i,0} + t\, x_{i,1},\ \sigma_x^2\, t(1-t)\, I_3\big), \quad \Omega_{i,t} \,|\, R_{i,0}, R_{i,1} \sim \mathcal{N}\big(t\,\omega_i,\ \sigma_R^2\, t(1-t)\, I_3\big),$$

with $R_{i,t} = R_{i,0} \exp(\widehat{\Omega_{i,t}})$ the pushforward of the tangent-space Gaussian.

A direct consequence of Proposition 2 is that the conditional drift on each modality decouples and is **identical to the deterministic SE(3) flow matching target of §3.2**. Differentiating the bridge SDEs in $\mathbb{R}^3$ and in $\mathfrak{so}(3)$ (Appendix A) yields

$$u^x_{i,t} \;=\; \frac{x_{i,1} - x_{i,t}}{1-t}, \qquad u^R_{i,t} \;=\; \frac{\log\!\big(R^\top_{i,t} R_{i,1}\big)^\vee}{1-t} \;\in\; T_{R_{i,t}} SO(3), \tag{7}$$

which is exactly (2)'s regression target. Conditional flow matching theory Lipman et al. [2022], Tong et al. [2024] guarantees that any pair of (legal) conditional sampler and conditional target supervises the corresponding marginal velocity field; ResBridge changes only the sampler in G_t, so the loss (2), the network v_θ, and the per-modality weights λ_x, λ_R are all reused verbatim. **ResBridge introduces no new trainable parameter and no new loss term.**

Centered translation prior. The flow matching backbone of §3.2 was trained on translation samples whose batch-residue mean is approximately zero (the OT-aligned prior of Yim et al. [2023a]). Injecting i.i.d. Gaussian noise into (3) breaks this calibration: the residue-mean of $\eta^x_i := \sigma_x \sqrt{t(1-t)}\, \varepsilon^x_i$ is $O(1/\sqrt{m_R})$ rather than zero, where m_R is the number of resolved residues per sample. We restore exact zero-mean translations by *post-multiply re-centering* the noise across resolved residues:

$$\tilde{\eta}^x_i \;=\; \eta^x_i \;-\; \frac{1}{m_R}\sum_{j\in\mathcal{R}} \eta^x_j, \qquad x_{i,t} \;=\; (1-t)x_{i,0} + t\, x_{i,1} + \tilde{\eta}^x_i, \tag{8}$$

so that $\sum_{i\in\mathcal{R}} \tilde{\eta}^x_i = 0$ holds bit-exact for every sample and v_θ continues to see corrupted batches whose translation distribution matches its training prior. Re-centering is sample-local and preserves the conditional Gaussian structure of (3) up to an $O(1/m_R)$ correction in the per-residue covariance, two orders of magnitude below the experimental B-factor noise floor (Appendix C). The same construction is applied transparently in §3.4, where it becomes essential under per-residue heteroscedasticity.

Sampling. At inference we replace the ODE Euler step of §3.2 by an Euler–Maruyama step on $SE(3)^L$. Given the current state $G_{i,t}$, the network produces a clean prediction $\hat{G}_{i,1} = (\hat{x}_{i,1}, \hat{R}_{i,1})$, the corresponding drift velocities $\hat{v}^x_{i,t} = (\hat{x}_{i,1} - x_{i,t})/(1-t)$ and $\hat{v}^R_{i,t} = \log(R^\top_{i,t}\hat{R}_{i,1})^\vee/(1-t)$, and the next state is

$$x_{i,t+\Delta t} = x_{i,t} + \hat{v}^x_{i,t}\,\Delta t + \sigma_x\sqrt{\Delta t}\,\xi^x_i, \qquad R_{i,t+\Delta t} = R_{i,t}\,\exp\!\Big(\,\hat{v}^R_{i,t}\,\Delta t + \widehat{\sigma_R}\sqrt{\Delta t}\,\xi^R_i\,\Big), \tag{9}$$

with $\xi^x_i, \xi^R_i \sim \mathcal{N}(0, I_3)$ independent across residues and timesteps. Iterating T steps from K independent initializations $G^{(k)}_0 \sim p_0$ produces a conformational ensemble $\{G^{(k)}_1\}_{k=1}^K$. We use $\sigma_x = 2.0$ Å and $\sigma_R = 0.25$ rad throughout, which match the typical magnitude of the drift residual that the deterministic backbone leaves at $t \approx 0.5$.

Strict containment of the deterministic backbone. Setting $\sigma_x = \sigma_R = 0$ collapses every Brownian bridge in (3)–(6) onto its mean and turns (9) into the deterministic ODE Euler step of §3.2. ResBridge therefore **strictly contains** the SE(3) flow matching backbone, both at training-time corruption and at inference-time integration. As a consequence, any pretrained backbone of the form (2) can be retrofitted to ResBridge by changing two scalar widths and a single line of the corruption / sampling loop, without retraining or fine-tuning the drift.

3.4 ResBridge-guided

B-factor as physical flexibility. Papers to be submitted to NeurIPS 2026 must be prepared according to the instructions presented here. Papers may only be up to **nine** pages long, including figures. **Papers that exceed the page limit will not be reviewed (or in any other way considered) for presentation at the conference.** Additional pages *containing acknowledgments, references, checklist, and optional technical appendices* do not count as content pages.

The margins in 2026 are the same as those in previous years.

Authors are required to use the NeurIPS LaTeX style files obtainable at the NeurIPS website as indicated below. Please make sure you use the current files and not previous versions. Tweaking the style files may be grounds for desk rejection.

3.5 Retrieval of style files

The style files for NeurIPS and other conference information are available on the website at

`https://neurips.cc`.

The only supported style file for NeurIPS 2026 is `neurips_2026.sty`, rewritten for LaTeX 2_ε. **Previous style files for LaTeX 2.09, Microsoft Word, and RTF are no longer supported.**

The LaTeX style file contains three optional arguments:

- `final`, which creates a camera-ready copy,
- `preprint`, which creates a preprint for submission to, e.g., arXiv,
- `nonatbib`, which will not load the `natbib` package for you in case of package clash.

At submission time, please omit the `final` and `preprint` options. This will anonymize your submission and add line numbers to aid review. Please do *not* refer to these line numbers in your paper as they will be removed during generation of camera-ready copies.

The file `neurips_2026.tex` may be used as a "shell" for writing your paper. All you have to do is replace the author, title, abstract, and text of the paper with your own.

4 General formatting instructions

The text must be confined within a rectangle 5.5 inches (33 picas) wide and 9 inches (54 picas) long. The left margin is 1.5 inch (9 picas). Use 10 point type with a vertical spacing (leading) of 11 points. Times New Roman is the preferred typeface throughout, and will be selected for you by default. Paragraphs are separated by ½ line space (5.5 points), with no indentation.

The paper title should be 17 point, initial caps/lower case, bold, centered between two horizontal rules. The top rule should be 4 points thick and the bottom rule should be 1 point thick. Allow ¼ inch space above and below the title to rules. All pages should start at 1 inch (6 picas) from the top of the page.

For the final version, authors' names are set in boldface, and each name is centered above the corresponding address. The lead author's name is to be listed first (left-most), and the co-authors' names (if different address) are set to follow. If there is only one co-author, list both author and co-author side by side.

5 Headings: first level

All headings should be lower case (except for first word and proper nouns), flush left, and bold.

First-level headings should be in 12-point type.

5.1 Headings: second level

Second-level headings should be in 10-point type.

5.1.1 Headings: third level

Third-level headings should be in 10-point type.

Paragraphs There is also a `\paragraph` command available, which sets the heading in bold, flush left, and inline with the text, with the heading followed by 1 em of space.

6 Citations, figures, tables, references

These instructions apply to everyone.

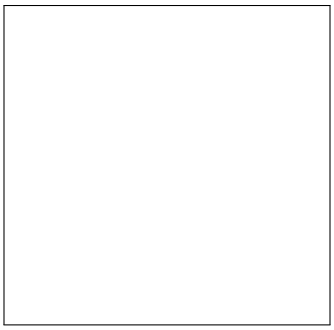

Figure 2: Sample figure caption. Explain what the figure shows and add a key take-away message to the caption.

6.1 Citations within the text

The `natbib` package will be loaded for you by default. Citations may be author/year or numeric, as long as you maintain internal consistency. As to the format of the references themselves, any style is acceptable as long as it is used consistently.

The documentation for `natbib` may be found at

`http://mirrors.ctan.org/macros/latex/contrib/natbib/natnotes.pdf`

Of note is the command `\citet`, which produces citations appropriate for use in inline text. For example,

```
\citet{hasselmo} investigated\dots
```

produces

Hasselmo, et al. (1995) investigated…

If you wish to load the `natbib` package with options, you may add the following before loading the `neurips_2026` package:

```
\PassOptionsToPackage{options}{natbib}
```

If `natbib` clashes with another package you load, you can add the optional argument `nonatbib` when loading the style file:

```
\usepackage[nonatbib]{neurips_2026}
```

6.2 Footnotes

Footnotes should be used sparingly. If you do require a footnote, indicate footnotes with a number[1] in the text. Place the footnotes at the bottom of the page on which they appear. Precede the footnote with a horizontal rule of 2 inches (12 picas).

Note that footnotes are properly typeset *after* punctuation marks.[2]

6.3 Figures

All artwork must be neat, clean, and legible. Lines should be dark enough for reproduction purposes. The figure number and caption always appear after the figure. Place one line space before the figure caption and one line space after the figure. The figure caption should be lower case (except for the first word and proper nouns); figures are numbered consecutively.

You may use color figures. However, it is best for the figure captions and the paper body to be legible if the paper is printed in either black/white or in color.

[1] Sample of the first footnote.
[2] As in this example.

Table 1: Sample table caption. Explain what the table shows and add a key take-away message to the caption.

	Part	
Name	Description	Size (μm)
Dendrite	Input terminal	$\approx$100
Axon	Output terminal	$\approx$10
Soma	Cell body	up to 10^6

6.4 Tables

All tables must be centered, neat, clean, and legible. The table number and title always appear before the table. See Table 1.

Place one line space before the table title, one line space after the table title, and one line space after the table. The table title must be lower case (except for the first word and proper nouns); tables are numbered consecutively.

Note that publication-quality tables *do not contain vertical rules*. We strongly suggest the use of the `booktabs` package, which allows for typesetting high-quality, professional tables:

`https://www.ctan.org/pkg/booktabs`

This package was used to typeset Table 1.

6.5 Math

6.6 Final instructions

Do not change any aspects of the formatting parameters in the style files. In particular, do not modify the width or length of the rectangle the text should fit into, and do not change font sizes. Please note that pages should be numbered.

7 Preparing PDF files

Fonts were the main cause of problems in the past years. Your PDF file must only contain Type 1 or Embedded TrueType fonts. Here are a few instructions to achieve this.

- You should directly generate PDF files using `pdflatex`.
- You can check which fonts a PDF files uses. In Acrobat Reader, select the menu Files>Document Properties>Fonts and select Show All Fonts. You can also use the program `pdffonts` which comes with `xpdf` and is
- The `\bbold` package almost always uses bitmap fonts. You should use the equivalent AMS Fonts:

  ```
  \usepackage{amsfonts}
  ```

  followed by, e.g., `\mathbb{R}`, `\mathbb{N}`, or `\mathbb{C}` for $\mathbb{R}$, $\mathbb{N}$ or $\mathbb{C}$. You can also use the following workaround for reals, natural and complex:

  ```
  \newcommand{\RR}{I\!\!R} %real numbers
  \newcommand{\Nat}{I\!\!N} %natural numbers
  \newcommand{\CC}{I\!\!\!\!C} %complex numbers
  ```

  Note that `amsfonts` is automatically loaded by the `amssymb` package.

If your file contains type 3 fonts or non embedded TrueType fonts, we will ask you to fix it.

7.1 Margins in LaTeX

Most of the margin problems come from figures positioned by hand using `\special` or other commands. We suggest using the command `\includegraphics` from the `graphicx` package. Always specify the figure width as a multiple of the line width as in the example below:

```
\usepackage[pdftex]{graphicx} ...
\includegraphics[width=0.8\linewidth]{myfile.pdf}
```

See Section 4.4 in the graphics bundle documentation (`http://mirrors.ctan.org/macros/latex/required/graphics/grfguide.pdf`)

A number of width problems arise when LaTeX cannot properly hyphenate a line. Please give LaTeX hyphenation hints using the `\-` command when necessary.

References

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Jason Yim, Brian L Trippe, Valentin De Bortoli, Emile Mathieu, Arnaud Doucet, Regina Barzilay, and Tommi Jaakkola. SE(3) diffusion model with application to protein backbone generation. In *International Conference on Machine Learning (ICML)*, 2023b.

A Pseudocode

B Technical appendices and supplementary material

Technical appendices with additional results, figures, graphs, and proofs may be submitted with the paper submission before the full submission deadline (see above). You can upload a ZIP file for videos or code, but do not upload a separate PDF file for the appendix. There is no page limit for the technical appendices.

Note: Think of the appendix as "optional reading" for reviewers. The paper must be able to stand alone without the appendix; for example, adding critical experiments that support the main claims to an appendix is inappropriate.

Algorithm 1. *Level 1: SE(3) flow matching*	**Algorithm 2.** *Level 2: ResBridge*	**Algorithm 3.** *Level 3: ResBridge-guided*
Training		
1. Sample $(s, G_1) \sim \mathcal{D}$, $G_0 \sim p_0$	1. Sample $(s, G_1) \sim \mathcal{D}$, $G_0 \sim p_0$	1. Sample $(s, G_1, B) \sim \mathcal{D}$, $G_0 \sim p_0$
2. $t \sim U(0,1)$; $G_t \leftarrow \mathrm{interp}(G_0, G_1, t)$	2. $t \sim U(0,1)$; $G_t \leftarrow \mathrm{interp}(G_0, G_1, t)$	2. $t \sim U(0,1)$; $G_t \leftarrow \mathrm{interp}(G_0, G_1, t)$
3. $\hat{G}_1 \leftarrow f_\theta(G_t, s, t)$	3. $\hat{G}_1,\ \mathbf{c} \leftarrow f_\theta(G_t, s, t)$	3. $\hat{G}_1,\ \mathbf{c} \leftarrow f_\theta(G_t, s, t)$
4. $\mathcal{L}_{\mathrm{drift}} \leftarrow \lVert G_1 - \hat{G}_1 \rVert^2$	4. $\mathcal{L}_{\mathrm{drift}} \leftarrow \lVert G_1 - \hat{G}_1 \rVert^2$	4. $\mathcal{L}_{\mathrm{drift}} \leftarrow \lVert G_1 - \hat{G}_1 \rVert^2$
	5. $r \leftarrow \mathrm{sg}(\lVert G_1 - \hat{G}_1 \rVert)$	5. $r \leftarrow \mathrm{sg}(\lVert G_1 - \hat{G}_1 \rVert)$
	6. $\alpha \leftarrow g_\phi(\mathbf{c}, G_t, \mathrm{sg}(\hat{G}_1), t)$	6. $\alpha \leftarrow g_\phi(\mathbf{c}, G_t, \mathrm{sg}(\hat{G}_1), t)$
	7. $\sigma \leftarrow \sigma_0(t) \cdot \alpha$	7. $\sigma \leftarrow \sigma_0(t) \cdot \alpha$
	8. $\mathcal{L}_{\mathrm{NLL}} \leftarrow r^2/\sigma^2 + \log \sigma^2$	8. $\mathcal{L}_{\mathrm{NLL}} \leftarrow r^2/\sigma^2 + \log \sigma^2$
		9. $\tilde{B} \leftarrow \mathrm{norm}_+(B)$
		10. $\mathcal{L}_{\mathrm{bf}} \leftarrow \lVert \alpha - \tilde{B} \rVert^2$
5. $\theta \leftarrow \mathrm{Adam}(\nabla_\theta \mathcal{L}_{\mathrm{drift}})$	9. $\theta \leftarrow \mathrm{Adam}(\nabla_\theta \mathcal{L}_{\mathrm{drift}})$	11. $\theta \leftarrow \mathrm{Adam}(\nabla_\theta \mathcal{L}_{\mathrm{drift}})$
	10. $\phi \leftarrow \mathrm{Adam}(\nabla_\phi \mathcal{L}_{\mathrm{NLL}})$	12. $\phi \leftarrow \mathrm{Adam}(\nabla_\phi(\mathcal{L}_{\mathrm{NLL}} + \lambda_B \mathcal{L}_{\mathrm{bf}}))$
Sampling		
1. $G_0 \sim p_0$	1. $G_0 \sim p_0$	1. $G_0 \sim p_0$
2. **for** $k = 0, \ldots, K-1$ **do**	2. **for** $k = 0, \ldots, K-1$ **do**	2. **for** $k = 0, \ldots, K-1$ **do**
3. $t \leftarrow k/K$	3. $t \leftarrow k/K$	3. $t \leftarrow k/K$
4. $\hat{G}_1 \leftarrow f_\theta(G_t, s, t)$	4. $\hat{G}_1,\ \mathbf{c} \leftarrow f_\theta(G_t, s, t)$	4. $\hat{G}_1,\ \mathbf{c} \leftarrow f_\theta(G_t, s, t)$
5. $v_t \leftarrow (\hat{G}_1 - G_t)/(1-t)$	5. $v_t \leftarrow (\hat{G}_1 - G_t)/(1-t)$	5. $v_t \leftarrow (\hat{G}_1 - G_t)/(1-t)$
	6. $\alpha \leftarrow g_\phi(\mathbf{c}, G_t, \hat{G}_1, t)$	6. $\alpha \leftarrow g_\phi(\mathbf{c}, G_t, \hat{G}_1, t)$
	7. $\sigma \leftarrow \sigma_0(t) \cdot \alpha;\ \xi \sim \mathcal{N}(0, I)$	7. $\sigma \leftarrow \sigma_0(t) \cdot \alpha;\ \xi \sim \mathcal{N}(0, I)$
6. $G_{t+\Delta t} \leftarrow G_t + v_t \Delta t$	8. $G_{t+\Delta t} \leftarrow G_t + v_t \Delta t + \sigma \sqrt{\Delta t}\, \xi$	8. $G_{t+\Delta t} \leftarrow G_t + v_t \Delta t + \sigma \sqrt{\Delta t}\, \xi$
7. **return** G_1	9. **return** G_1	9. **return** G_1

Figure 3: **The three levels of ResBridge, aligned row by row.** A single three-line table; each row corresponds to one conceptual operation, so identical lines line up horizontally and a level that does not contain a given operation simply leaves its cell blank. Blue marks every line (or sub-expression) that Level 2 introduces on top of Level 1 (the heteroscedastic stochastic bridge); red marks every line (or sub-expression) that Level 3 introduces on top of Level 2 (the B-factor anchor). Both colors persist across columns so that the layering remains visible at a glance. *Notation.* f_θ: drift FlowModel, returning the predicted clean state $\hat{G}_1$ and (in Level 2/3) its single/pair embeddings $\mathbf{c}$; g_ϕ: BridgeWidthNet, returning a per-residue bounded modulation $\alpha \in [\alpha_{\min}, \alpha_{\max}]$; $\sigma_0(t) = \lambda\sqrt{t(1-t)}$: endpoint-safe base schedule; σ: per-residue bridge width; $\mathrm{sg}(\cdot)$: stop-gradient, decoupling drift gradients from g_ϕ; norm_+: log-shift-scale positive normalization of B-factors. The drift parameters θ and the bridge parameters ϕ are updated by *separate* optimizers, which is what keeps the bridge purely an "observer" of the drift.

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